

Software Requirement Specification(SRS)

RentRide - Online Vehicle Rental and Booking Management System

1. Introduction

1.1 Purpose

The purpose of this document is to specify the requirements for the Online Vehicle Rental and Booking Management System. The system automates vehicle rental operations by allowing customers to register, browse available vehicles, book vehicles online, view booking history, and manage rentals while enabling administrators to manage vehicles, customers, returns, and system operations.

1.2 Scope

The Online Vehicle Rental and Booking Management System is a web-based application developed using React for the frontend, Spring Boot for the backend, and MySQL as the database.

The system provides:

- Customer registration and login
- Vehicle browsing and search
- Vehicle availability checking
- Online vehicle booking
- Automatic booking confirmation
- Automatic bill generation
- Vehicle return management
- Customer booking history
- Dashboard

The application aims to eliminate manual record keeping, prevent double bookings, and improve operational efficiency for vehicle rental businesses.

1.3 Objectives

- Provide a centralized vehicle rental platform.
- Enable customers to reserve vehicles online.
- Prevent duplicate bookings.
- Maintain vehicle availability records.
- Automate billing processes.
- Improve customer experience.
- Reduce manual administrative work.

1.4 Definitions

Term	Description
Customer	Registered user who rents vehicles
Admin	System administrator
Booking	Reservation of a vehicle for a specific duration
Rental Period	Duration between pickup and return dates
Vehicle Status	Current availability status of a vehicle

2. Overall Description

2.1 Product Perspective

The system acts as a centralized platform connecting customers and vehicle rental administrators.

Customers can:

- Register and login
- Browse available vehicles
- Search vehicles
- Book vehicles
- Cancel bookings
- View booking history
- View rental bills

Administrators can:

- Manage vehicles
- Manage customers
- View bookings
- Process vehicle returns
- Add damage charges
- View dashboard and revenue reports

2.2 Product Functions

Customer Functions

- Registration
- Login
- Search vehicles
- Check vehicle availability
- Book vehicles

- Cancel bookings
- View booking history
- View bills
- Update profile

Administrator Functions

- Login
- Manage vehicles
- Manage customers
- View bookings
- Process vehicle returns
- Add damage charges
- View dashboard

2.3 User Classes

Administrator

Responsibilities:

- Manage system operations
- Manage vehicle inventory
- Monitor bookings
- Process returns
- View reports

Customer

Responsibilities:

- Register account
- Book vehicles
- Track bookings
- View bills

2.4 Assumptions

- Internet connection is available.
- Customers possess valid driving licenses.
- Vehicles are available only if not already booked during selected dates.
- Customers provide accurate registration information.

3. Functional Requirements

FR-1 User Registration

Description

The system shall allow customers to create an account.

Inputs

- Name
- Email
- Phone Number
- Password
- Driving License Number

Outputs

- Successful registration message

FR-2 User Login

Description

The system shall authenticate registered users.

Inputs

- Email
- Password

Outputs

- Access to user dashboard

FR-3 Vehicle Management

Description

The system shall allow administrators to manage vehicle records.

Features

- Add vehicle
- Update vehicle
- Delete vehicle
- View vehicle details

Vehicle Information

- Vehicle Number
- Vehicle Name
- Brand
- Type
- Rent Per Day
- Status
- Vehicle Image

FR-4 Vehicle Search

Description

The system shall allow customers to search available vehicles.

Search Filters

- Vehicle Type
- Brand
- Rental Price
- Availability

FR-5 Vehicle Availability Check

Description

The system shall verify vehicle availability before booking.

Validation

The booking shall not be allowed if the vehicle is already booked during the selected rental period.

FR-6 Booking Management

Description

The system shall allow customers to book available vehicles automatically.

Inputs

- Vehicle Selection
- Pickup Date
- Return Date

Process

1. Customer selects vehicle.
2. Customer enters rental dates.
3. System checks vehicle availability.
4. System calculates rental amount.
5. Booking is confirmed automatically.

Outputs

- Booking ID
- Booking Confirmation
- Rental Cost

Booking Status

- BOOKED

- ACTIVE
- COMPLETED
- CANCELLED

FR-7 Booking History

Description

The system shall allow customers to view their booking history.

Information Displayed

- Booking ID
- Vehicle Name
- Booking Date
- Pickup Date
- Return Date
- Rental Amount
- Booking Status

FR-8 Return Management

Description

The administrator shall process vehicle returns.

Inputs

- Booking ID
- Actual Return Date
- Damage Charges (if applicable)

Outputs

- Return Confirmation
- Updated Bill

FR-9 Billing Management

Description

The system shall automatically generate bills.

Formula

Total Amount =
(Rent Per Day × Number of Rental Days)

- Damage Charges

Outputs

- Bill Number
- Booking ID
- Customer Name
- Vehicle Details
- Total Amount

FR-10 Dashboard

Description

The system shall display system statistics.

Statistics

- Total Vehicles
- Available Vehicles
- Active Rentals
- Completed Rentals
- Total Customers

4. Non-Functional Requirements

4.1 Performance

- System response time shall be less than 3 seconds.
- Search results shall be displayed within 2 seconds.

4.2 Security

- Passwords shall be encrypted using BCrypt.
- JWT authentication shall be implemented.
- Unauthorized users shall not access protected resources.
- Role-based access control shall be implemented.

4.3 Reliability

- System shall maintain data consistency.
- Booking records shall not be lost.
- Duplicate bookings shall be prevented.

4.4 Availability

- System shall be available 24/7.

4.5 Scalability

- New vehicle categories can be added.
- Additional user roles can be introduced.
- Payment integration can be added in future.

4.6 Usability

- User-friendly interface.
- Easy navigation.
- Responsive design for desktop and mobile devices.

5. System Requirements

Software Requirements

Frontend

- React.js
- HTML
- CSS
- JavaScript

Backend

- Java
- Spring Boot
- Spring Security
- Spring Data JPA

Database

- MySQL

Testing Tools

- Postman

Development Tools

- IntelliJ IDEA
- VS Code
- MySQL Workbench

Hardware Requirements

Server

- Processor: Intel i3 or above
- RAM: 4GB minimum

- Storage: 20GB

Client

- Modern Web Browser
- Internet Connection

6. Use Case Specifications

Use Case 1: Customer Registration

Actor

Customer

Flow

1. Open registration page.
2. Enter personal information.
3. Submit registration form.
4. System validates data.
5. Account is created successfully.

Use Case 2: Vehicle Booking

Actor

Customer

Flow

1. Login.
2. Browse available vehicles.
3. Select vehicle.
4. Select pickup and return dates.
5. System verifies availability.
6. System calculates rental amount.
7. Booking is confirmed automatically.

Use Case 3: View Booking History

Actor

Customer

Flow

1. Login.
2. Navigate to booking history.
3. View current and previous bookings.

Use Case 4: Vehicle Return

Actor

Administrator

Flow

1. Open active booking.
2. Mark vehicle as returned.
3. Enter damage charges if applicable.
4. System updates final bill.
5. Vehicle becomes available for future bookings.

7. Database Design

Users

Field	Type
id	Long
name	String
email	String
password	String
phone	String
licenseNumber	String
role	String

Vehicles

Field	Type
id	Long
vehicleName	String
vehicleNumber	String

brand	String
type	String
rentPerDay	Double
status	String
imageUrl	String

Bookings

Field	Type
id	Long
userId	Long
vehicleId	Long
bookingDate	Date
pickupDate	Date
returnDate	Date
totalAmount	Double
status	String

Bills

Field	Type
id	Long
bookingId	Long
amount	Double
damageCharges	Double
generatedDate	Date

8. System Architecture

Frontend Layer

React Application

Responsibilities:

- User Interface
- Form Validation
- API Requests

Backend Layer

Spring Boot Application

Responsibilities:

- Authentication
- Vehicle Management
- Booking Logic
- Availability Checking
- Billing Logic

Database Layer

MySQL Database

Responsibilities:

- Data Storage
- Relationship Management
- Query Processing

9. Future Enhancements

- Online Payment Gateway
- Vehicle Image Upload
- Email Notifications
- SMS Notifications
- GPS Tracking
- Vehicle Maintenance Management
- Customer Reviews and Ratings
- Mobile Application
- QR-Based Vehicle Pickup Verification

10. Conclusion

The Online Vehicle Rental and Booking Management System provides an efficient platform for managing vehicle rentals. The system automates customer registration, vehicle booking, availability checking, billing, and return management. It improves operational efficiency, prevents booking conflicts, enhances customer experience, and provides a scalable solution for vehicle rental businesses.